Unit 1-Part 1

Thinking Power

What is Intelligence?

- There is not yet any formal definition of intelligence.
- Intelligence is the computational part of the ability to achieve goals in the world.
- Varying kinds and degrees of intelligence occur in people, many animals and some machines.
- We cannot yet characterize in general what kinds of computational procedures we want to call intelligent.

What is AI?

MAN MADE

- Artificial Intelligence is the study of how to make computers do things at which at the moment people do better.
- It is the science and engineering of making intelligent machines, especially intelligent computer programs.
- The term AI was first introduced by John McCarthy in 1960.

Calculate, Learn, Reason, Decide, Adapt.

Why AI?

There are several limitations of human intelligence (HI).

- Unreliable: The quality and quantity of HI varies from time to time, space to space due to physical and mental exertion of human beings.
- Limited Applications: Te HI cannot be always employed due to cost and lack of expertise or non availability of expertise at particular time or geographical position.
- Speed: The speed with which HI is able to react to situations to exhibit intelligence is much slower than the AI systems.
- Failure proof: HI is not failure proof but AI system has only hardware failure which is rare.

AI – compromise between the versatility and limitations of natural intelligence.

In standard AI taxonomy (specifically Stuart Russell and Peter Norvig's *Artificial Intelligence: A Modern Approach* framework), AI is split into four distinct quadrants:

- **Acting Humanly:** Behaving like a human (e.g., passing the Turing Test, imitating conversation).
- **Thinking Humanly:** Replicating human cognitive processes and brain mechanics (Cognitive Science).
- **Thinking Rationally:** Following pure rules of logic and correct inference ("Laws of Thought").
- **Acting Rationally:** Taking actions that maximize the chance of achieving a specific goal given available information (**The Rational Agent Approach**).

Most modern AI systems focus entirely on **Acting Rationally**—making the optimal decision to solve a problem—rather than trying to emulate human emotions, fallacies, or biological quirks.

look at what modern AI systems do every second: **Predict, Classify, Generate, Recommend.** Which of these four actually requires *thinking*? Is prediction the same as intelligence? A barometer predicts rain based on atmospheric pressure drops. Is a barometer intelligent? No. So prediction alone cannot be our definition of intelligence.

An **intelligent AI system** is an autonomous entity that perceives its environment through sensors, processes information to learn and reason, and takes actions that maximize its chances of successfully achieving a specific goal—even in complex, dynamic, or uncertain situations.

In technical terms (specifically the standard **Rational Agent** framework in computer science), intelligence isn't measured by whether a system has a human-like mind, consciousness, or biological feelings. It is measured by its **problem-solving performance and adaptivity**.

The 5 Defining Capabilities

To be considered genuinely intelligent rather than just automated, a system must exhibit five key traits:

- **Perception:** Interprets unstructured environment inputs (vision, text, audio, sensor telemetry) into structured context.
- **Rational Decision-Making:** Weighs multiple possibilities to select the statistically optimal choice given available data.
- **Adaptation & Learning:** Improves performance over time without needing manual code rewrites when given new experiences or data.
- **Handling Uncertainty:** Acts effectively when information is incomplete, noisy, or continuously changing.
- **Goal Alignment:** Directs internal capabilities toward achieving an objective (e.g., winning a game, minimizing casualties, discovering a drug compound).

WHY AI

- With the help of AI, you can create such software or devices which can solve real-world problems very easily and with accuracy such as health issues, marketing, traffic issues, etc.
- With the help of AI, you can create your personal virtual Assistant, such as Google Assistant, Siri, etc.
- With the help of AI, you can build such Robots which can work in an environment where survival of humans can be at risk.
- AI opens a path for other new technologies, new devices, and new Opportunities.

AI System Examples	Primary Goal	Core AI Paradigm	Key Example
Multimodal Frontier	Cross-modal understanding	Transformers & Latent Embeddings	Gemini 2.5 Pro, GPT-5
Agentic Frameworks	Autonomous execution	ReAct (Reason + Act) Loops	CrewAI, Moveworks
Physical Robotics	Spatial navigation & control	Vision-Language-Action (VLA)	Waymo, Figure 02
Scientific AI	Modeling physical laws	Graph Neural Nets & Diffusion	AlphaFold 3

AI Approaches

- Discipline that systematizes and automates reasoning processes to create machines that:

Think like humans	Think rationally
Act like humans	Act rationally

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Thinking Humanly: Cognitive Modelling

- This method tries to copy how humans think and understand things. It aims to understand how humans think and solve problems. i.e. Cognitive science.
- "Cognitive science/modelling" refers to the scientific study of how humans think and reason.

- • **Example:** Cognitive psychology-inspired AI systems, such as virtual assistants like Siri or Alexa, try to act like humans. They listen to what people say, try to understand what they mean, and respond like a person would in a conversation.
- • It is not related to how a computer program solves a problem correctly. Thinking Humanly is more interested in seeing how its steps compare to how a human would solve the same problem.

Acting Humanly: The Turing Test

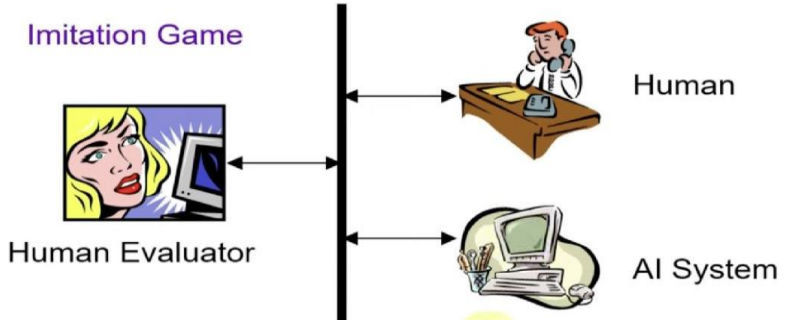
- This approach aims to create AI systems that perform tasks in a manner very similar to human behaviour. It focuses on achieving human-like performance in various tasks.
- **Example:** Turing Test is a classic example of the acting humanly approach. In the Turing Test, a human evaluator interacts with a machine and a human through a text interface without knowing which is which. If the evaluator cannot reliably distinguish between the machine and the human based on the conversation, the machine is considered to have passed the test.

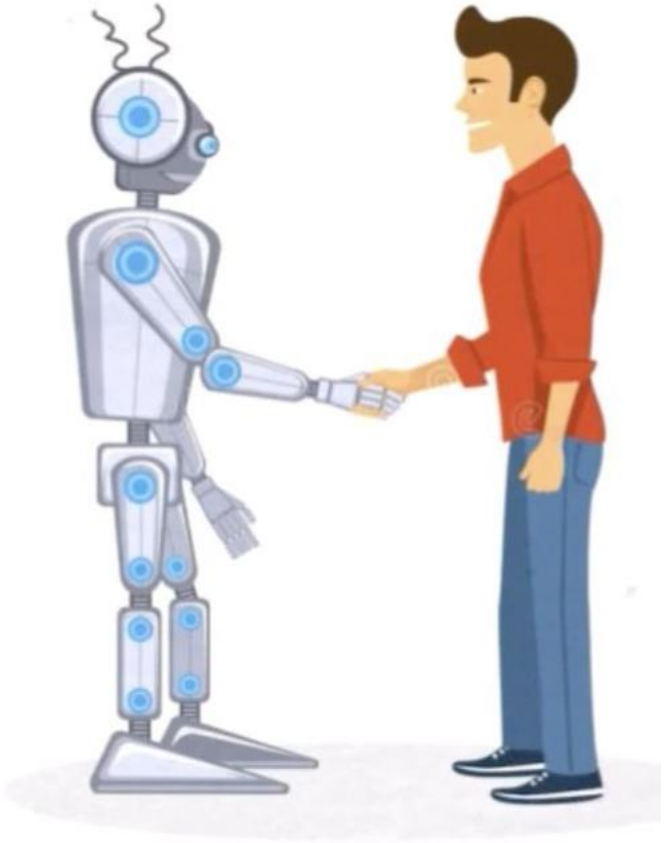
❖ Acting Humanly: The Turing Test

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Alan Turing (1912-1954) performed a operational test for intelligent behavior: the Imitation(Copying) Game, "Computing Machinery and Intelligence" (1950)

Predicted that by 2000, a machine might have a 30% chance of fooling a normal person for 5 minutes.



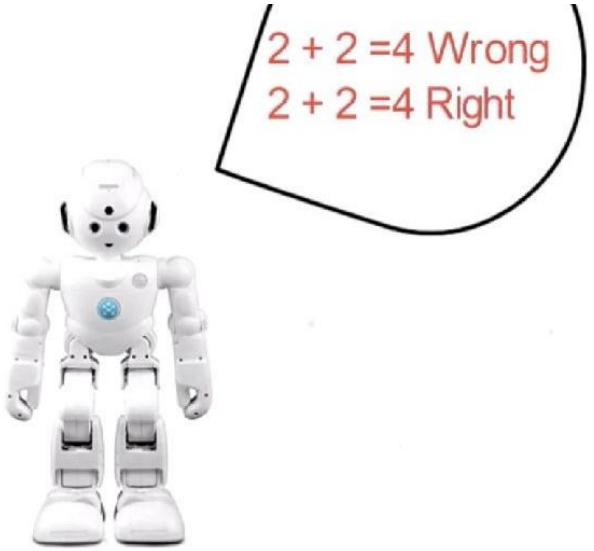


Thinking Rationally: Laws of Thought

- This approach emphasizes on designing AI systems that follow principles of logic and rational(logical) decision-making, regardless of whether it mirrors human thought processes.
- In simple words, if your thoughts are based on facts and not emotions, it is called rational thinking.
- **Example:** Expert systems, such as medical diagnosis systems, rely on rules and logical inference to make decisions. These systems use knowledge bases of medical expertise and logical reasoning to diagnose diseases based on symptoms reported by patients.

Thinking Rationally

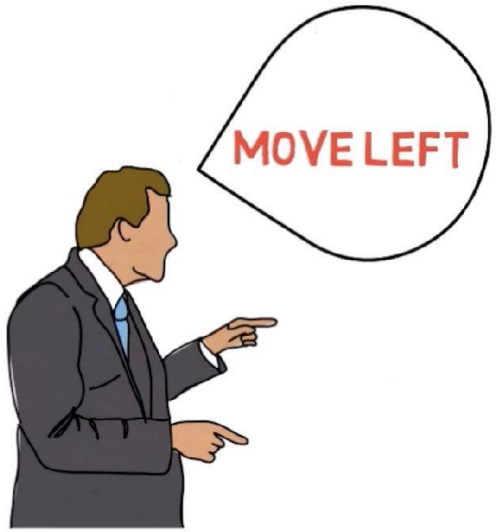
This approach study about the computations that make it possible to perceive, reason, and act of machine in a right manner.



Acting Rationally

- •This approach focuses on creating AI systems that make decisions to achieve the best outcome, irrespective of whether the decision-making process resembles human thinking.
- Acting rationally is more related to scientific development than human-based approaches.
- **Example:** Autonomous vehicles, like self-driving cars, operate based on the principle of acting rationally. They analyze sensor data in real-time, process it using algorithms to detect obstacles, and make decisions to navigate safely to their destination, optimizing factors like speed, distance, and safety.

- How can a machine act correctly ?
- This approach says if we design intelligents behavior in agents, than we make a machine which act reasonly and correctly .







Type of Tasks

- *Task 1: Computers outperform human beings:* data storing and retrieval, calculations, making large number of copies. The general feature of this kind is the routine jobs that requires minimum amount of intelligence.

- *Task 2: Human beings outperform computers:* solving ambiguity, finding similarities among dissimilarities and vice versa, making sense of relative importance, intuition, i.e., finding the solution to a problem only by feeling and without any reasoning (the sixth sense).

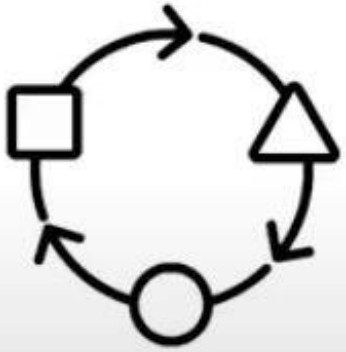
- AI – Building machines which can perform task 2 as well.

What Comprises to Artificial Intelligence?

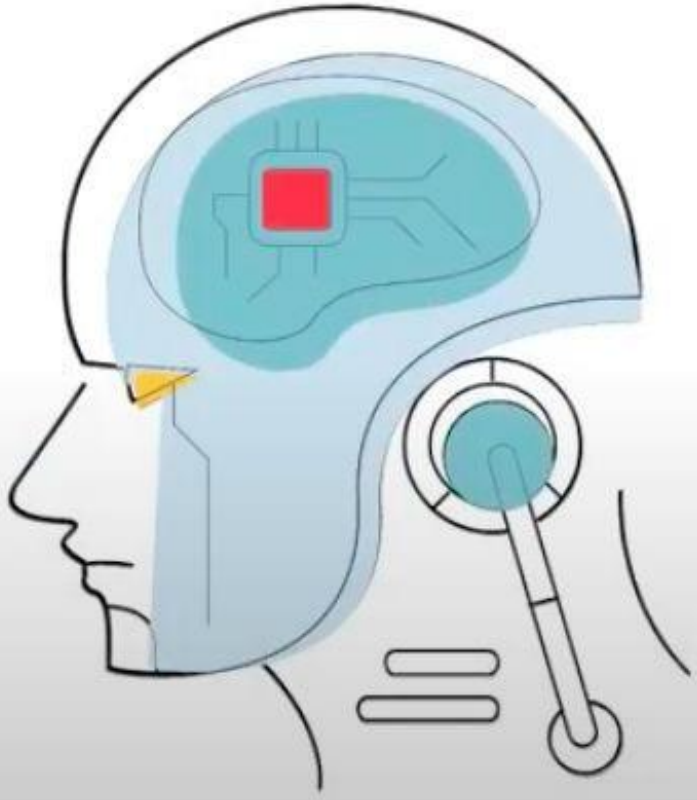
- Artificial Intelligence is not just a part of computer science even it's so vast and requires lots of other factors which can contribute to it. To create the AI first we should know that how intelligence is composed, so the Intelligence is an intangible part of our brain which is a combination of **Reasoning, learning, problem-solving perception, language understanding, etc.**

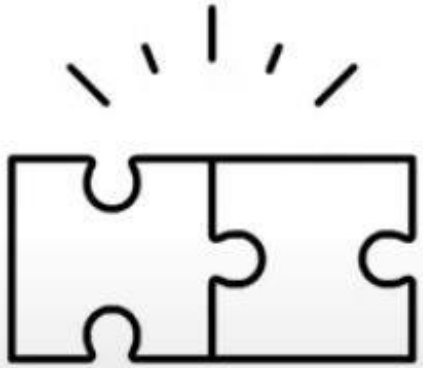


REASON



ADAPT





PROVIDE SOLUTIONS

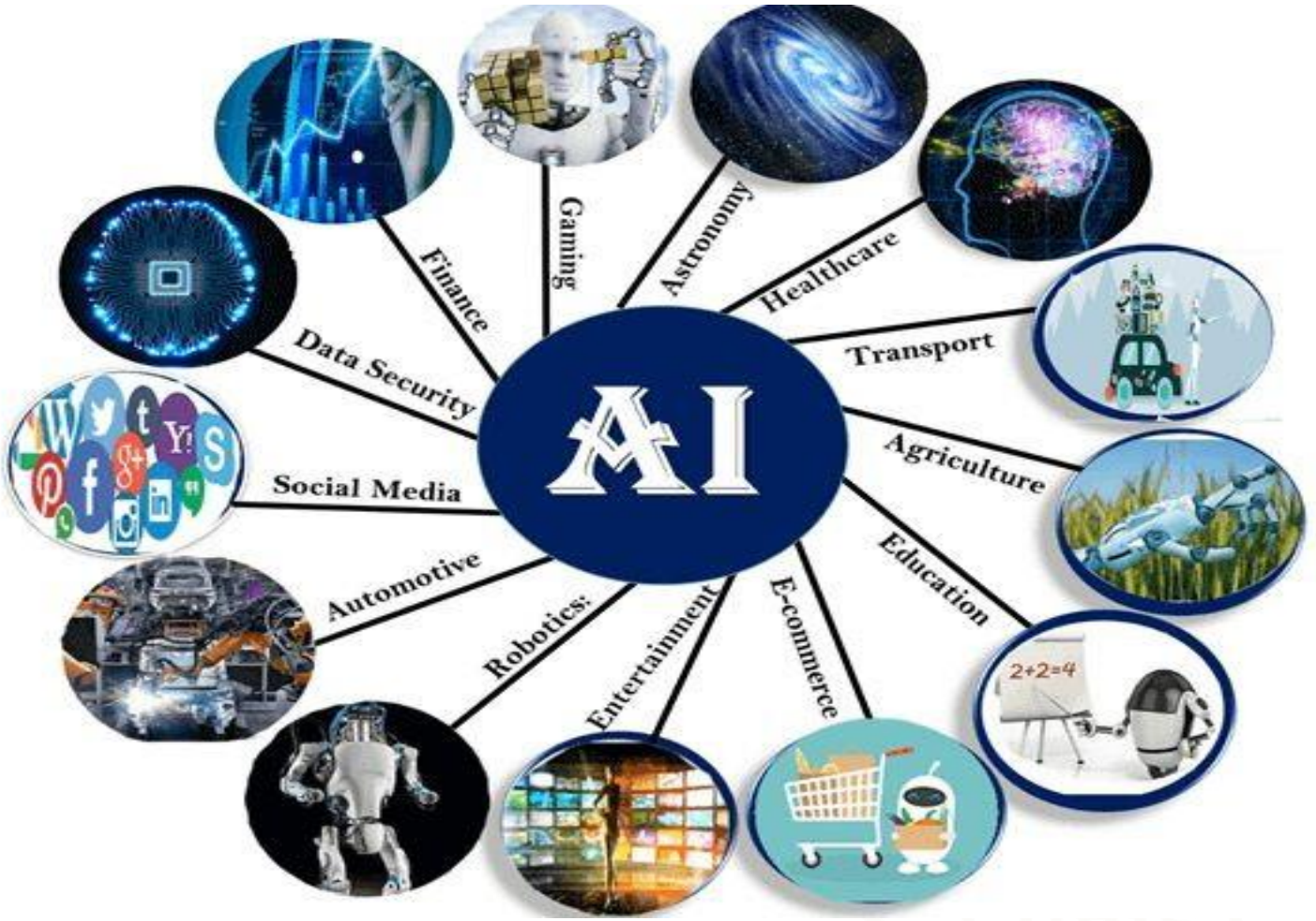
Advantages of Artificial Intelligence

- High Accuracy with less errors: AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.
- High-Speed: AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
- High reliability: AI machines are highly reliable and can perform the same action multiple times with high accuracy.
- Useful for risky areas: AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.
- Digital Assistant: AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.
- Useful as a public utility: AI can be very useful for public utilities such as a self-driving car which can make our journey safer and hassle-free, facial recognition for security purpose, Natural language processing to communicate with the human in human-language, etc

Disadvantages of Artificial Intelligence

- **High Cost:** The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
- **Can't think out of the box:** Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.
- **No feelings and emotions:** AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.
- **Increase dependency on machines:** With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.
- **No Original Creativity:** As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

Application of AI



- 1. AI in Astronomy

Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

- 2. AI in Healthcare

In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.

Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

- 3. AI in Gaming

AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

4. AI in Finance

AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

5. AI in Data Security

The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

6. AI in Social Media

Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyse lots of data to identify the latest trends, hashtag, and requirement of different users.

7. AI in Travel & Transport

AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

8. AI in Automotive Industry

Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.

Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

- 9. AI in Robotics:

Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.

Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

- 10. AI in Entertainment

We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

11. AI in Agriculture

Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.

12. AI in E-commerce

AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color, or even brand.

13. AI in education:

AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant.

AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

Turing Test in Artificial Intelligence

The Turing Test is a widely recognized benchmark for evaluating a machine's ability to demonstrate human-like intelligence. The core idea is simple: A human judge engages in a text-based conversation with both a human and a machine. The judge's task is to determine which participant is human and which is the machine. If the judge is unable to distinguish between the human and the machine based solely on the conversation, the machine is said to have passed the Turing Test.

Criteria for the Turing Test

The Turing Test does not require the machine to be correct or logical in its responses but rather to be convincing in simulating human conversation. The test is fundamentally about deception—the machine must fool the judge into believing that it is human.

The key criteria include the following for turing test:

- **Natural Language Processing (NLP):** The machine must understand and generate human language fluently.
- **Knowledge Representation:** The machine needs to handle and manipulate knowledge to provide contextually relevant responses.
- **Reasoning:** The machine should demonstrate some form of logical reasoning, even if flawed, to sustain a conversation.
- **Learning:** Ideally, the machine should learn from the interaction, adapting its responses over time.

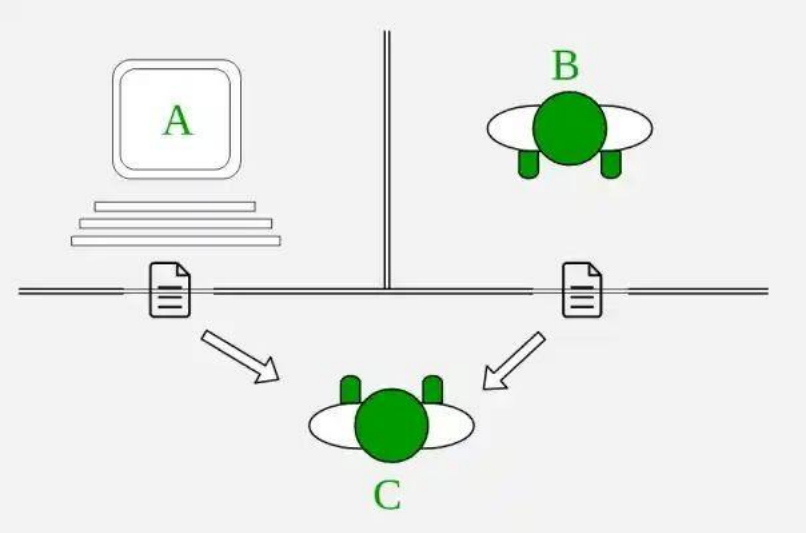
- **1. Natural Language Processing**
- The machine must be able to **understand and respond in human language.**
- This means it should understand questions, grammar, sentence meaning, and context. It should also generate answers that sound natural and human-like.
- **Example:**
  If a person asks, "How was your day?" the machine should not only understand the words but also respond in a conversational way, such as, "It was good. I learned something new today."
- So, Natural Language Processing helps the machine communicate with humans smoothly.

- **2. Knowledge Representation**
- The machine should be able to **store and use knowledge about the world.**
- It must know facts, concepts, relationships, and situations so that it can give meaningful answers. Knowledge representation allows the AI to organize information in a form that the computer can understand and use.
- **Example:**
  The machine should know that "a doctor works in a hospital," "Paris is the capital of France," or "rain can make roads slippery."
- Without knowledge representation, the machine may give random or incomplete answers.

- **3. Reasoning**
- The machine should be able to **think logically and draw conclusions** from the information it has.
- Reasoning helps AI solve problems, answer complex questions, and make decisions. It should not only repeat stored information but also use logic.
- **Example:**
  If the machine knows:
  "All humans need oxygen."
  "Ravi is a human."
  Then it should conclude:
  "Ravi needs oxygen."
- This shows that the machine can connect facts and come to a logical answer.

- **4. Learning**
- The machine should be able to **learn from experience and improve over time.**
- This means it should adapt based on previous interactions, mistakes, or new information. A learning machine becomes better at answering questions and understanding users.
- **Example:**
  If a chatbot gives a wrong answer and later receives correction, it should remember the correction and avoid the same mistake in the future.
- Learning makes the machine more intelligent and useful over time.

- **How the Turing Test Works?**
- In a typical Turing Test scenario, three participants are involved: two humans and one machine.
- The interrogator, a human judge, is isolated from the other two participants. The judge asks questions to both the human and the machine, aiming to identify which one is the human. The machine's goal is to respond in a way that makes it indistinguishable from the human participant. If the judge cannot reliably identify the machine, the machine is considered to have passed the Turing Test


A
B
C

Advantages of the Turing Test in Artificial Intelligence

- **Evaluating Machine Intelligence:** The Turing Test provides a simple and well-known method for assessing machine intelligence.
- **Setting a Benchmark:** It establishes a benchmark for AI research and offers a goal for researchers to strive towards.
- **Inspiring Research:** The Turing Test has inspired numerous studies and experiments aimed at developing machines that can pass the test, driving progress in AI.
- **Simple to Administer:** The Turing Test is relatively easy to administer, requiring just a computer and a human judge.

Disadvantages of the Turing Test in Artificial Intelligence

- **Limited Scope:** The Turing Test focuses primarily on language-based conversations and does not account for other aspects of intelligence, such as perception, problem-solving, and decision-making.
- **Human Bias:** The results can be influenced by the biases and preferences of the human judge, making it difficult to obtain objective and reliable results.
- **Not Representative of Real-World AI:** The Turing Test may not accurately represent the kind of intelligence that machines need to demonstrate in real-world applications.